

Roll: 220105

GitHub Profile: <https://github.com/mdraihaan27>

GitHub Repository:

<https://github.com/mdraihaan27/neural-network>

Collab File Link:

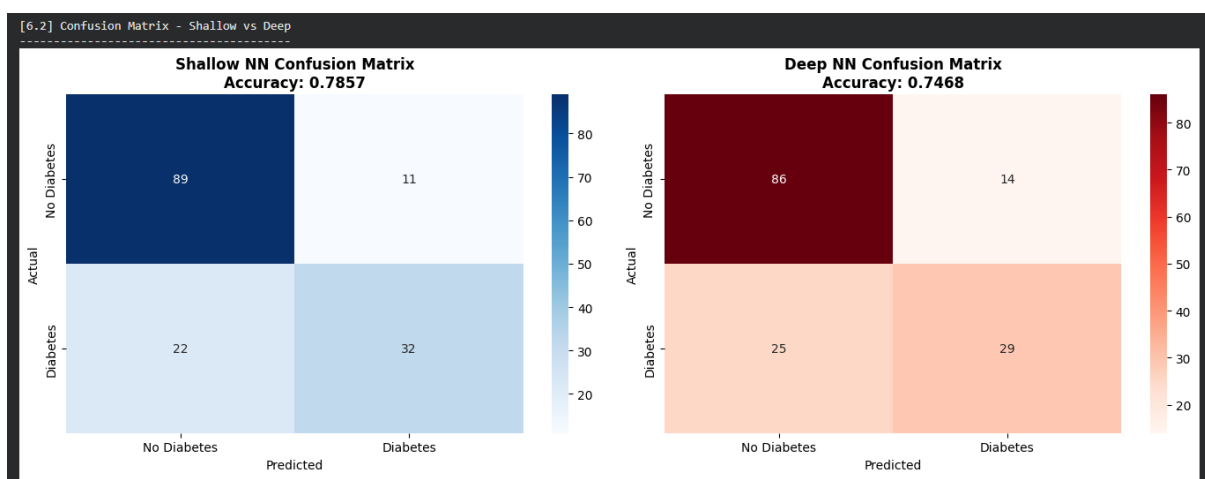
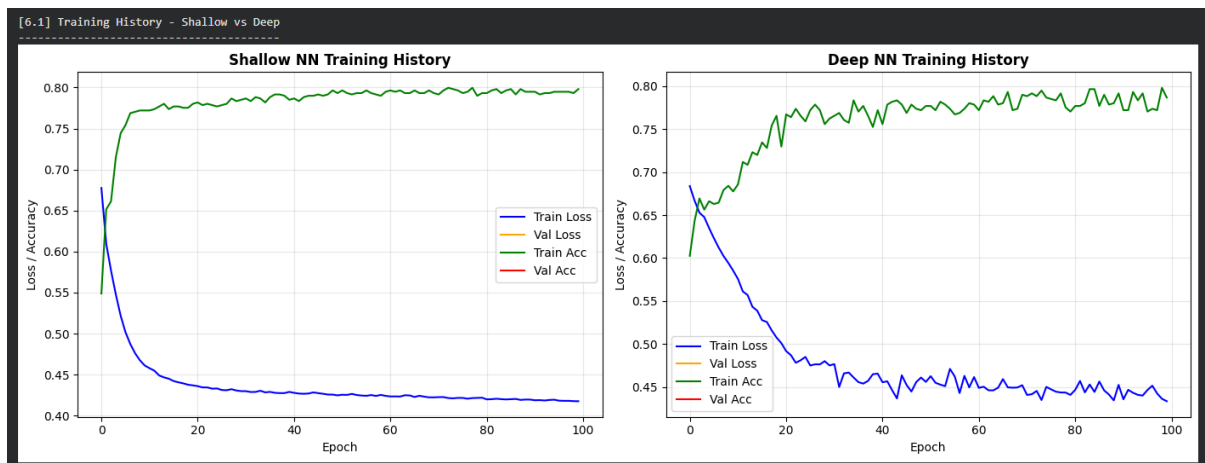
https://colab.research.google.com/drive/1cAm3_xGQeNMRaUgG7XEJxefc53z5xjwQ?usp=s_haring

Dataset Link:

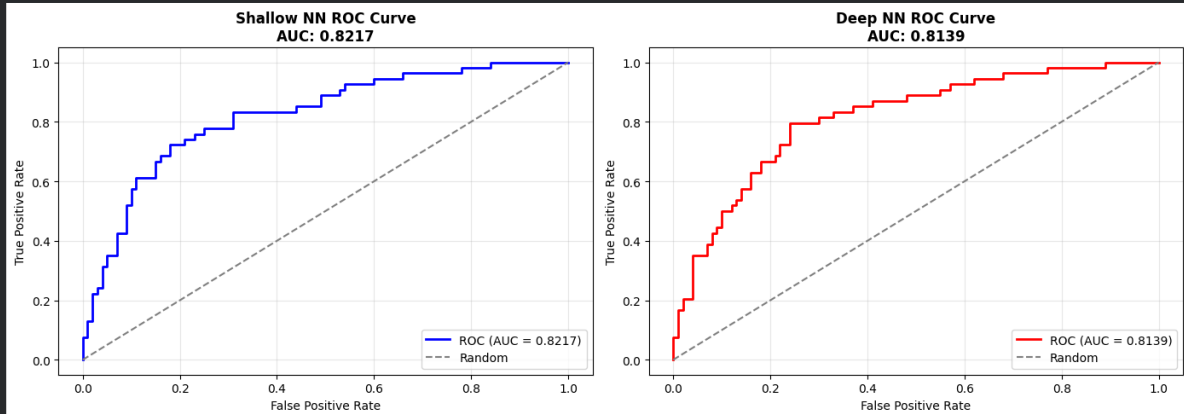
[GitHub link](#)

[Kaggle link](#)

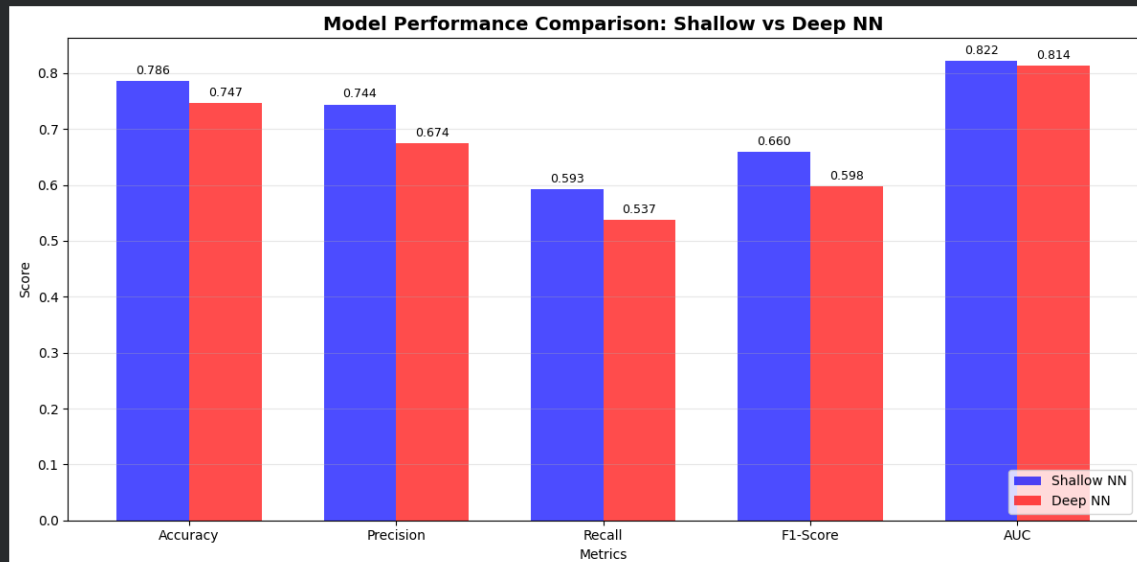
Screenshots:



[6.3] ROC Curve - Shallow vs Deep



[6.4] Evaluation Metrics - Combined Bar Chart



SHALLOW NEURAL NETWORK ARCHITECTURE

Input Layer: 21 features

Hidden Layer 1: 64 neurons, Activation: sigmoid

Output Layer: 1 neuron, Activation: Sigmoid

Total Parameters:

- fc1.weight: 1,344 parameters
- fc1.bias: 64 parameters
- fc2.weight: 64 parameters
- fc2.bias: 1 parameters
- Total: 1,473 trainable parameters

DEEP NEURAL NETWORK ARCHITECTURE

Input Layer: 21 features

Hidden Layer 1: 64 neurons, Activation: tanh, Dropout: 0.3

Hidden Layer 2: 64 neurons, Activation: tanh, Dropout: 0.3

Hidden Layer 3: 64 neurons, Activation: tanh, Dropout: 0.3

Output Layer: 1 neuron, Activation: Sigmoid

Total Parameters:

- fc1.weight: 1,344 parameters
- fc1.bias: 64 parameters
- fc2.weight: 4,096 parameters
- fc2.bias: 64 parameters
- fc3.weight: 4,096 parameters
- fc3.bias: 64 parameters
- fc4.weight: 64 parameters
- fc4.bias: 1 parameters
- Total: 9,793 trainable parameters

[6.5] Network Structure - Architecture Summary

SHALLOW NEURAL NETWORK ARCHITECTURE

Input Layer: 21 features
Hidden Layer 1: 64 neurons, Activation: sigmoid
Output Layer: 1 neuron, Activation: Sigmoid
Total Parameters:
- fc1.weight: 1,344 parameters
- fc1.bias: 64 parameters
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- Total: 1,473 trainable parameters

DEEP NEURAL NETWORK ARCHITECTURE

Input Layer: 21 features
Hidden Layer 1: 64 neurons, Activation: tanh, Dropout: 0.3
Hidden Layer 2: 64 neurons, Activation: tanh, Dropout: 0.3
Hidden Layer 3: 64 neurons, Activation: tanh, Dropout: 0.3
Output Layer: 1 neuron, Activation: Sigmoid
Total Parameters:
- fc1.weight: 1,344 parameters
- fc1.bias: 64 parameters
- fc2.weight: 4,096 parameters
- fc2.bias: 64 parameters
- fc3.weight: 4,096 parameters
- fc3.bias: 64 parameters
- fc4.weight: 64 parameters
- fc4.bias: 1 parameters
- Total: 9,793 trainable parameters

ARCHITECTURE COMPARISON TABLE

Feature	Shallow NN	Deep NN
Number of Layers	1 Hidden Layer	3 Hidden Layers
Hidden Units	64	64
Activation	sigmoid	tanh
Regularization	None	Dropout (0.3)
Optimizer	Adam	sgd
Learning Rate	0.001	0.001
Total Parameters	1,473	9,793